

AI Museum of the Future

Weekly Validation Report: Corpus, Chunking, Retrieval, Agents, Prompts and Multimodal Components

Esprit 2026-2027, 5DS, Advanced Deep Learning (AI Automation Final Project). October 6, 2026.

Summary

This report documents the validation of six milestone items for the retrieval-augmented generation and multi-agent pipeline developed this week. Rather than relying on unit testing alone, the team validated the complete system by running it end to end against production infrastructure, specifically a live Supabase vector database and the OpenAI API. This approach revealed seven integration defects that unit tests could not have detected, since those tests mock the very interfaces at which the defects occurred. Each item discussed below is therefore supported by evidence obtained through direct execution of the system, rather than by design intent alone.

Milestone item	Status	Evidence
Corpus collected	Complete	Over 6,700 records across ten or more sources, normalized through one canonical pipeline into Supabase
Chunking strategy implemented	Complete	Fixed-size and recursive strategies, both covered by property-based tests
Initial retrieval tests completed	Complete	Five queries executed against the live corpus, discussed in Section 3
Agents implemented and communicating	Complete	Five specialist agents and a curator agent produced a complete, grounded exhibition during a live run
Prompts V1 defined and tested	Complete	Versioned prompt library, including a documented revision described in Section 5
Multimodal component working in isolation	Complete	A real photograph processed into a structured description and an embedding using GPT-4o vision

1. Corpus Collection

Data collection is organized as a single canonical pipeline fed by two complementary sets of collectors, both drawing exclusively on official public application programming interfaces; no content was obtained through web scraping.

The first set of collectors focuses on Tunisia and draws on six sources: GDELT, Wikipedia (through both the pageviews endpoint and REST article summaries), Wikidata (queried via SPARQL), Wikimedia Commons, arXiv, and the World Bank Open Data API, producing 2,613 normalized records spanning 2020 to 2026.

The second set of collectors gathers material across a broader range of global and topical sources: GDELT, arXiv, the Metropolitan Museum of Art Open Access API, Openverse, the Internet Archive, and DBpedia. It spans thirty-six countries, including all twenty-two Arab League member states, and more than twenty-two topics ranging from artificial intelligence and climate change to culture, the economy, sport, and the 2026 FIFA World Cup, yielding over 3,750 further records and more than 700 downloaded images.

Both sets of collectors are normalized into the same record schema and written to the same processed-data directory, from which a single ingestion step chunks, embeds, and stores every record in the shared Supabase project. GDELТ and arXiv records from either side share the same source label and merge naturally; Openverse, the Metropolitan Museum of Art, the Internet Archive, and DBpedia contribute four sources the Tunisia-focused collectors do not cover. A third contribution to the same shared corpus came from the team's n8n automation workflow, which satisfies the project's automation platform requirement and independently ingested 153 documents from arXiv, NewsAPI, and BBC and Reuters open data feeds.

The selection rationale for each source, together with known limitations such as rate limiting, coverage gaps, and formatting inconsistencies, is documented in `data/README.md`. The combined corpus across all three contributions exceeds 6,700 records; ingestion of the complete set into the shared Supabase project was under way at the time of writing, with a snapshot of 2,767 documents and 2,777 chunks confirmed and queryable.

2. Chunking Strategy

The module `rag/chunking/chunker.py` implements two chunking strategies and allows either to be selected at runtime, as required by the assignment. The fixed-size strategy slides a token window, tokenized with `tiktoken`, across the input text with configurable overlap. The recursive strategy splits text first on paragraph boundaries, then on lines, falling back to fixed-size splitting for fragments that remain too large.

Both strategies are covered by unit tests and by a property-based test, implemented with Hypothesis, that verifies non-empty output is produced for any non-empty input text across one hundred randomly generated examples per run.

3. Retrieval Evaluation

The module `rag/vector_store/retriever.py` embeds an incoming query using OpenAI's `text-embedding-3-small` model, issues a `match_chunks` remote procedure call to Postgres that performs cosine similarity search through `pgvector`, and returns ranked results with an optional reranking pass using `gpt-4o`. Five queries were executed against the complete, live corpus to evaluate retrieval quality.

Query	Top similarity	Top result (abridged)
key AI breakthroughs in 2026	0.591	"The rise of AI over the last 8 decades..." (Wikimedia)
Tunisia economy and unemployment	0.390	GDP per capita indicator for the North Africa region (World Bank)
climate change protests and activism	0.467	"Street art in Sydney. Climate change. 2026" (Wikimedia)
contemporary art in North Africa	0.546	"Zeitz Museum of Contemporary Art Africa, Cape Town" (Wikimedia)
space exploration milestones	0.285	Relevant arXiv research on robotics and world models

A genuine defect was identified while preparing this evaluation. The vector index built on the chunks table, an ivfflat index from `pgvector`, had been constructed with parameters unsuited to the table's row count, with the effect that it silently returned zero results for every query and raised no error. This is a documented `pgvector` pitfall. The index was diagnosed and rebuilt with corrected parameters during this session, after which retrieval returned the consistently relevant results shown above.

4. Multi-Agent Synthesis

Five specialist agents, namely Historian, Sociologist, Technology, Culture and Art, and Visual, independently analyze the same retrieved evidence from complementary perspectives, sharing their findings through a cross-reference layer. A curator agent then synthesizes the five perspectives into a structured exhibition consisting of a title, a concept statement, between four and six narrative sections, and a list of selected artifacts.

A complete live run of the pipeline, executed through `test_agents_live.py`, produced an exhibition titled “Voices of Change: Technology, Culture, and Activism in 2026,” comprising five grounded sections (Advancements in Artificial Intelligence, Quantum Computing Milestones, Global Climate Activism, Cultural Expressions of Activism, and Visual Language of Change) together with a list of selected artifacts drawn directly from the retrieved evidence.

5. Prompt Engineering

A versioned prompt library, held in `prompts/*.md`, covers all five specialist agents together with the curator agent, and is evaluated through `prompts/evaluate_prompts.py`. A concrete revision was carried out and documented during this session.

In its initial form, the curator prompt's example output showed only two sections and did not state the system's actual requirement of four to six sections anywhere in its instructions. During the live run, the pipeline failed validation with the error “curator output must include four to six sections,” as the model had mirrored the structure of the example rather than the constraint enforced by the code.

The revised prompt states the requirement explicitly within the instructions and extends the example to five placeholder sections rather than two. When re-run against the identical input that had previously failed, the revised prompt produced a valid result immediately.

6. Multimodal Analysis

The module `multimodal/analyze_image.py` submits an image to GPT-4o vision and extracts a structured description consisting of detected objects, a scene summary, a free-text description, and thematic tags, together with a 1536-dimension embedding intended for later retrieval. The component was tested against real photographs drawn from the collected corpus, for example a news image obtained through GDELT, and produced accurate, well-structured output.

A further defect was identified and corrected in this component. The original parsing logic assumed that the model would respond using a flat label-and-value format, whereas the model in practice replies using Markdown headings such as “#### Objects.” This mismatch caused the parser to raise an exception, which was silently caught and replaced with a generic fallback message, discarding a perfectly usable description in the process. The parsing logic was rewritten to extract fields defensively, without raising, regardless of the exact formatting the model chooses to use.

7. Implementation Challenges and Resolutions

Running the complete pipeline against production infrastructure, rather than treating a passing unit test suite of 247 tests as sufficient evidence of correctness, surfaced seven integration defects that mocked tests could not have revealed. Each was diagnosed, corrected, verified, and committed within the same session, and is summarized in Table 2.

No.	Observed symptom	Root cause	Resolution
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1	Retrieval silently returned zero rows, with no error raised	The ivfflat vector index had been built with parameters unsuited to the table's row count	The index was rebuilt with corrected parameters and now returns relevant results
2	Document insertion failed with a schema error	The application code assumed a source_id column that had never been added to the schema	The column was added, together with a unique index to support proper deduplication
3	Re-running ingestion either failed or produced duplicate rows	The upsert operations did not specify a conflict target	An explicit on_conflict clause was added to both the document and chunk upsert operations
4	Ingestion failed on a single Wikidata record	A bare-year date value such as "2013" is rejected outright by a timestamp column	A defensive date parser was introduced; values that cannot be parsed are retained in metadata instead
5	A multi-thousand-record ingestion run terminated after one network interruption	No retry logic existed around the database calls	A retry mechanism with backoff was added around both upsert operations
6	The curator agent failed output validation	The prompt did not state its own requirement of four to six sections	The prompt was revised, as described in Section 5
7	Multimodal analysis silently returned "image unavailable"	The parser failed on the model's actual Markdown-formatted response	The extraction logic was rewritten to be defensive and non-raising

Each resolution was verified against its originally failing case and against the complete automated test suite before being committed and pushed to the shared repository.

8. Current State and Future Work

At the time of writing, the automated test suite passes in full, with 247 of 247 tests succeeding across unit and property-based categories. Data collection, normalization, chunking, embedding, and storage now run as a single canonical pipeline, described in Section 1, rather than as separate unconnected efforts. The shared Supabase project is being populated with the full combined corpus of more than 6,700 records as a result. Three scripts were prepared for live demonstration during validation: `demo_retrieval.py`, `test_agents_live.py`, and `demo_multimodal.py`.

Future work includes extending the n8n automation workflow to call the same canonical ingestion path directly, so that all three collection efforts converge not only on a shared schema and database but on a single orchestrated run, and incorporating the LLM-based reranking pass already implemented in the retriever into the default retrieval path once its latency and cost trade-offs have been evaluated against the project's constraints.